Math Exam 07 — Chapter 8: Geometry of Straight Lines

Seemab | Class 9 SSC-I (FBISE / NBF Federal Board) | Attempt: 5 Jun 2026 | Marked against Version 1 Answer Key

TOTAL SCORE	GRADE	SECTION A (MCQ)	SECTION B (SHORT)	SECTION C (LONG)	MCQ STREAK
64/65	**A+**	**11/12**	**33/33**	**20/20**	**11**
98.5%	Distinction	Q2 wrong	Perfect	Perfect	streak ends here*

SECTION A — Multiple Choice Questions (12 Marks)

#	Concept	Seemab	Correct	Result	Note
1	Definition of gradient	A	**A**	✓	rise ÷ run correctly identified
2	Gradient of a horizontal line	**D**	**C**	✗	Chose D (gradient = -1); correct is C (gradient = 0). A horizontal line has inclination 0°, so m = tan 0° = 0.
3	Condition for perpendicular lines	D	**D**	✓	m1 × m2 = -1 correctly identified
4	Slope-intercept form	A	**A**	✓	y = mx + c correctly identified
5	Slope from general form 6x - 5y + 15 = 0	A	**A**	✓	Slope = -a/b = 6/5 correctly found
6	Line parallel to y-axis through (3, 5)	B	**B**	✓	x = 3 correctly identified
7	x-intercept of x + y = 5	A	**A**	✓	Set y = 0 gives x = 5
8	Angle between two lines formula	A	**A**	✓	tanθ = (m2 - m1)/(1 + m1m2) correctly identified
9	Condition for three collinear points	B	**B**	✓	Equal slopes of any two segments
10	Normal form: meaning of p	C	**C**	✓	Length of perpendicular from origin
11	Equation y = 0 represents	B	**B**	✓	x-axis correctly identified
12	Intersection of 9x - 7y = 0 and 8x - 11y = 0	A	**A**	✓	Both pass through origin (0, 0)

Section A: 11 / 12 | One deduction: Q2 (horizontal gradient = -1 chosen instead of 0)

SECTION B — Short Questions (33 Marks)

Attempt all 11 pairs

Q.2(i) OR: Find inclination of line with slope -1.732 — 3 / 3

Variant chosen: OR (inclination from m = -1.732)

m = tanθ = -1.732 stated. θ = tan⁻¹(-1.732). Reference angle is 60°, adjusted to 2nd quadrant: θ = 180° - 60° = 120°. All three marking points earned.

Correctly applied the 2nd-quadrant rule; reference angle method clean and explicit.

Q.2(ii) Main: gradient and inclination of line joining A(2,6) and B(5,8) — 3 / 3

Variant chosen: Main

m = (8-6)/(5-2) = 2/3. θ = tan⁻¹(2/3) ≈ 33.69°. All three steps correct.

Q.2(iii) OR: Find x if slope of line through A(3, x) and B(5, 8) is 4 — 3 / 3

Variant chosen: OR

m = (8-x)/(5-3) = 4. Cross-multiply: 8-x = 8. x = 0. Correct setup, correct algebra, correct answer.

Q.2(iv) Main: Prove X(0,-3), Y(4,7), Z(6,12) collinear using slopes — 3 / 3

Variant chosen: Main

Slope XY = (7-(-3))/(4-0) = 10/4 = 5/2. Slope YZ = (12-7)/(6-4) = 5/2. Also verified slope XZ = 15/6 = 5/2. Since all slopes equal, points are collinear. Full proof with conclusion stated.

Computed all three slopes as additional verification — beyond minimum requirement. Good habit.

Q.2(v) OR: Equation of line through (-5, 7) with slope 4 — 3 / 3

Variant chosen: OR

Point-slope form set up correctly. y - 7 = 4(x + 5). Expanded: y - 4x = 27. Rearranged to 4x - y + 27 = 0. All correct.

Q.2(vi) Main: Equation through (2,4) and (-1,-4) using two-point form **3 / 3**

Variant chosen: Main

$(y-4)/(-8) = (x-2)/(-3)$. Cross multiply: $-3(y-4) = -8(x-2)$. Expanded and rearranged to $8x - 3y - 4 = 0$. All three marks earned.

Q.2(vii) OR: Reduce $4x - 3y + 9 = 0$ to two-intercept form **3 / 3**

Variant chosen: OR

$4x - 3y = -9$. Divided to get $x/(-9/4) + y/3 = 1$. x-intercept = $-9/4$, y-intercept = 3. Correct.

Q.2(viii) Main: Line through (3,-2) perpendicular to line with slope 3 **3 / 3**

Variant chosen: Main

$m_1 = 3$ so perpendicular slope $m_2 = -1/3$. $y + 2 = -1/3(x - 3)$. Multiplied out: $3y + 6 = -(x-3)$. Final: $x + 3y + 3 = 0$. Fully correct.

Q.2(ix) Main: Angle from l_1 to l_2 when $m_1 = -1$, $m_2 = 2$ **3 / 3**

Variant chosen: Main

$\tan\theta = (2-(-1))/(1+(2)(-1)) = 3/(-1) = -3$. $\theta = \tan^{-1}(-3)$. Reference angle 71.5°, so inclination = 180° - 71.5° = 108.5°. (Key answer 108.43° — rounding difference only.) Full credit.

Q.2(x) Main: Point of intersection of $2x + y + 1 = 0$ and $x - y - 4 = 0$ **3 / 3**

Variant chosen: Main

Added equations: $3x - 3 = 0$, $x = 1$. Substituted to get $y = -3$. Point (1, -3) correct.

Q.2(xi) Main: Line through intersection of $x + 2y = 0$ and $x - y = 3$, passing through (1,1) **3 / 3**

Variant chosen: Main

Family of lines: $(x + 2y) + k(x - y - 3) = 0$. Substituted (1,1): $3 - 3k = 0$, $k = 1$. Combined equation: $2x + y - 3 = 0$. All steps correct and clearly shown.

Family-of-lines method handled correctly; k substituted and verified cleanly.

Section B: 33 / 33 | Perfect — all 11 questions fully correct

SECTION C — Long Questions (20 Marks) Attempt all 4 questions

Q.3(i) OR: A(-2,1), B(6,3), C(10,5), D(2,3) — prove ABCD is a parallelogram **5 / 5**

Variant chosen: OR (parallelogram proof)

All four slopes computed correctly: Slope AB = 2/8 = 1/4. Slope BC = 2/4 = 1/2. Slope CD = -2/-8 = 1/4. Slope AD = 2/4 = 1/2. Conclusion: AB ∥ CD (slope 1/4) and BC ∥ AD (slope 1/2). Both pairs of opposite sides parallel, hence ABCD is a parallelogram.

Clean systematic layout: all four slopes computed then paired for parallel-sides conclusion. Full 5/5.

Q.3(ii) Main: Reduce $6x - 5y + 15 = 0$ into all six standard forms **5 / 5**

Variant chosen: Main (all six forms)

(i) Slope-intercept: $y = (6/5)x + 3$ — correct.
(ii) Two-intercept: $x/(-5/2) + y/3 = 1$ — correct.
(iii) Point-slope: $y - 3 = (6/5)(x - 0)$ using point (0, 3) — correct.
(iv) Two-point: $(y-3)/(0-3) = (x-0)/(-5/2-0)$ — correct.
(v) Normal form: Rewrote as $-6x + 5y = 15$, divided by $\sqrt{61}$: $(-6/\sqrt{61})x + (5/\sqrt{61})y = 15/\sqrt{61}$. Calculation of $\sqrt{36+25} = \sqrt{61}$ shown explicitly — correct.
(vi) Symmetric form: $(x-0)/(5/\sqrt{61}) = (y-3)/(6/\sqrt{61})$ — correct.

All six forms derived. The normal form working including the $\sqrt{61}$ Pythagorean check was shown on the side of the page — good discipline.

Q.3(iii) Main: Interior angles of triangle ABC: A(-2,0), B(3,0), C(6,5) **5 / 5**

Variant chosen: Main

Slopes: $m(AB) = 0$, $m(BC) = 5/3$, $m(AC) = 5/8$ — all correct.
Angle at A: $\tan\alpha = (5/8 - 0)/(1 + 0) = 5/8$, $\alpha = \tan^{-1}(5/8) \approx 32°$.
Angle at B: $\tan\beta = (5/3 - 0)/(1 + 0) = 5/3$, $\beta = \tan^{-1}(5/3) \approx 59°$.
Angle at C: $\gamma = 180° - 32° - 59° = 89°$.
Verification: 32° + 59° + 89° = 180° — explicitly shown and confirmed.

Verification step done spontaneously — the 180° check adds a mark and shows rigorous working habit. Full 5/5.

Q.3(iv) OR: Nasir's fruit sales (fruiters Rs.120/dozen, Shakri Malta Rs.150/dozen, earned Rs.1200) **5 / 5**

Variant chosen: OR (real-world straight line application)

(i) Standard form: $120x + 150y = 1200$, divided by 30 gives $4x + 5y = 40$. Correct setup with explicit division step shown.
(ii) $y = 4$ dozen Shakri Malta: $4x + 20 = 40$, $4x = 20$, $x = 5$ dozens. Correct.
(iii) Slope-intercept: $5y = -4x + 40$, $y = -(4/5)x + 8$. Slope = -4/5 interpreted correctly on page 19: "for each additional dozen of fruiters sold, Shakri Malta dozens decrease by 4/5 (fixed budget trade-off)." Full verbal interpretation given.

The verbal interpretation of the slope in part (iii) is exactly what the mark scheme requires and was written clearly on the final page. Full 5/5.

Section C: 20 / 20 | Perfect — all 4 long questions fully correct

STRENGTHS — THIS EXAM

- Section B and C: flawless — not a single mark dropped across 44 marks of written work
- Family-of-lines method (Q2xi): k substituted and verified cleanly; first time this appeared on an exam
- All six standard forms in Q3(ii): normal form √61 Pythagorean calculation shown explicitly
- Verification habit: Q3(iii) angle sum = 180° spontaneously confirmed; Q2(iv) computed all three slopes beyond the minimum required
- Applied maths (Q3iv OR): verbal slope interpretation written out correctly — often skipped
- 2nd-quadrant inclination rule: correctly applied in Q2(i) OR and Q2(ix)

AREA FOR REVISION — THIS EXAM

- **Q2 MCQ:** Gradient of a horizontal line — chose D ($m = -1$) instead of C ($m = 0$). A horizontal line has inclination 0°, so $m = \tan 0° = 0$. The gradient -1 belongs to a line at 135°. Flash-card: horizontal = gradient 0, vertical = gradient undefined.
- This is the first MCQ error since Biology exam-01 in April (ending an 11-exam perfect MCQ streak at 11).

Math Exam History (this subject)

Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
18 Apr 2026	exam-01-ch1-2	12/12	33/33	20/20	65/65	100%	A+
22 Apr 2026	exam-03-ch4-5-7	12/12	28.5/33	20/20	60.5/65	93.1%	A+
25 May 2026	exam-04-ch4	12/12	26/33	19/20	57/65	87.7%	A
5 Jun 2026	**exam-07-ch8 (today)**	**11/12**	**33/33**	**20/20**	**64/65**	**98.5%**	**A+**

Section B has fully recovered (33/33 today vs 26/33 in the previous exam). Section C is back to perfect. The only mark dropped was one MCQ, representing a significant rebound from the prior exam's Section B deductions.